

3838. Weighted Word Mapping

Difficulty: Easy

Link: <https://leetcode.com/problems/weighted-word-mapping/?envType=daily-question&envId=2026-06-13>

Generated at: 2026-06-13 11:27:34

Problem:

You are given an array of strings `words`, where each string represents a word containing lowercase English letters.

You are also given an integer array `weights` of length 26, where `weights[i]` represents the weight of the i^{th} lowercase English letter.

The **weight** of a word is defined as the **sum** of the weights of its characters.

For each word, take its weight modulo 26 and map the result to a lowercase English letter using reverse alphabetical order (0 -> 'z', 1 -> 'y', ..., 25 -> 'a').

Return a string formed by concatenating the mapped characters for all words in order.

Example 1:

Input: words = ["abcd","def","xyz"], weights = [5,3,12,14,1,2,3,2,10,6,6,9,7,8,7,10,8,9,6,9,9,8,3,7,7,2]

Output: "rij"

Explanation:

- The weight of "abcd" is $5 + 3 + 12 + 14 = 34$. The result modulo 26 is $34 \% 26 = 8$, which maps to 'r'.
- The weight of "def" is $14 + 1 + 2 = 17$. The result modulo 26 is $17 \% 26 = 17$, which maps to 'i'.
- The weight of "xyz" is $7 + 7 + 2 = 16$. The result modulo 26 is $16 \% 26 = 16$, which maps to 'j'.

Thus, the string formed by concatenating the mapped characters is "rij" .

Example 2:

Input: words = ["a","b","c"], weights = [1,1]

Output: "yyy"

Explanation:

Each word has weight 1. The result modulo 26 is $1 \% 26 = 1$, which maps to 'y'.

Thus, the string formed by concatenating the mapped characters is "yyy" .

Example 3:

Input: words = ["abcd"], weights = [7,5,3,4,3,5,4,9,4,2,2,7,10,2,5,10,6,1,2,2,4,1,3,4,4,5]

Output: "g"

Explanation:

The weight of "abcd" is $7 + 5 + 3 + 4 = 19$. The result modulo 26 is $19 \% 26 = 19$, which maps to 'g'.

Thus, the string formed by concatenating the mapped characters is "g" .

Constraints:

- `1 <= words.length <= 100`
- `1 <= words[i].length <= 10`
- `weights.length == 26`
- `1 <= weights[i] <= 100`
- `words[i]` consists of lowercase English letters.